Mathematics Applications and Interpretations
Internal Assessment (HL)

Topic:

OPTIMIZATION OF SURFACE AREA AND VOLUME TO ESTIMATE THE COST OF THE CERAMIC VASE.

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Introduction and Objective –

Recently, I had a conversation with my Uncle, who works in the ceramics industry, and he showed me how they make ceramic vases like the one in Figure 1, (Citation 1). His company produces vases with various unique shapes, which made me curious about the math involved in their creation. I started thinking about how the surface area and volume of a vase influence the amount of material needed and whether the vase's design could be optimized to reduce manufacturing costs.

The vase I have is 6 inches tall with a base diameter of 3.5 inches. I considered how to model this mathematically and realized that calculating the surface area and volume of revolution would be ideal for this purpose. By optimizing the vase's surface area and volume, it might be possible to use less material, which would be especially helpful for large-scale production.

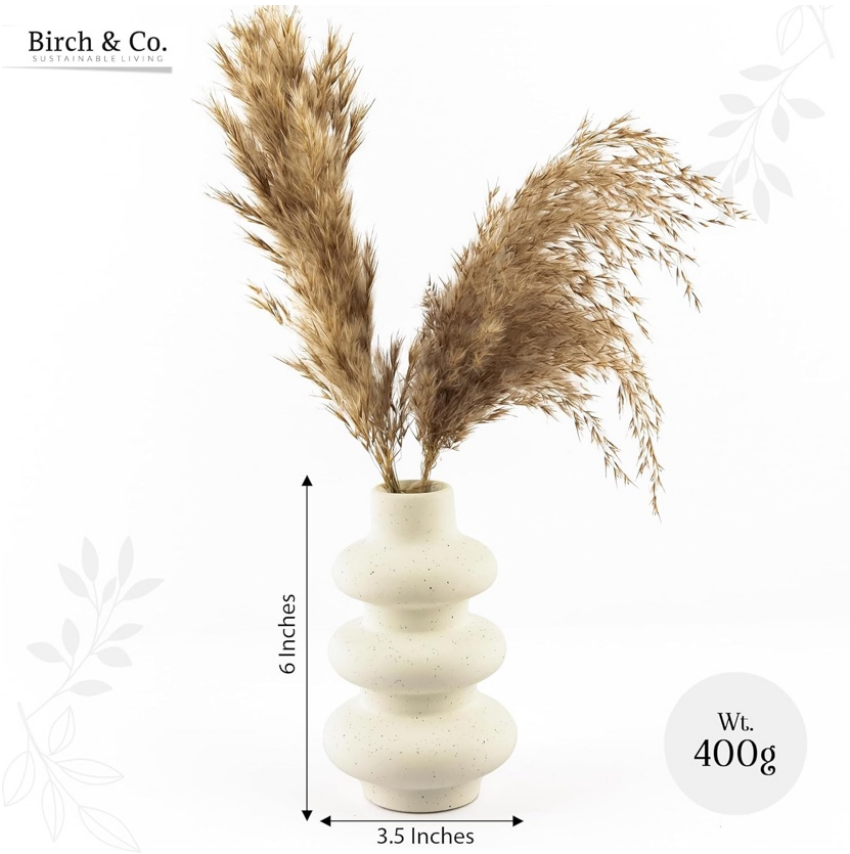

Figure 1: Ceramic Vase (Company website)
(Citation 1)

Thus, I decided to explore how to mathematically model the vase's surface area and volume to lower the material used in the production per vase. My goal is to estimate the material costs, particularly the volume of ceramic needed for production and then reduce it. By optimizing the volume, and keeping the design the same to the maintain the aesthetic appeal of the vase, I aim to reduce the cost and produce more number of vases in a given amount of material, especially for companies like my uncle's that produce similar vases on a large scale.

Figure 2: Ceramic pot at my place

So, I have to –

1) Find the functions of the ceramic vase.
2) Calculate the External volume of the ceramic vase using the method of solids of revolution.
3) Calculate the Internal volume of the ceramic vase by checking the amount of water inside.
4) Determine the surface area of the vase using integration.
5) Analyze the cost of ceramic and number of vases produced in a particular amount of ceramic material.
6) Optimize the vase's design to minimize surface area and volume, ensuring material efficiency without sacrificing its appearance.
7) Design the new optimized vase, to check for any inaccuracies and better understanding.
8) Calculate the External and Internal volume, and surface area of new optimized vase.
9) Analyze and compare the cost of ceramic and number of optimized vases produced in amount of ceramic material.

Finding the functions of ceramic vase.

Looking at Figure 2, to find the functions of vase I will divide the vase into two parts, section 1 and section 2 (figure 3). Section 1 (red) consists of a series of even horizontal edges, which I have to determine. Section 2 (blue) consists of a series of parabolic curves whose expressions are yet to be found. Therefore, to find the volume, I need coordinates of each unique curve and vertical edge, on the vase. As shown in figure 3, there are 11 coordinates from (A to K), and from these I can determine the equations for both the vertical edges (1, 3, 5, 7) and for the inverse parabola curves (2, 4, 6). The curve and vertical edges would revolve around the x – axis and generate the volume and surface area. The x – axis the axis of symmetry (figure 3).

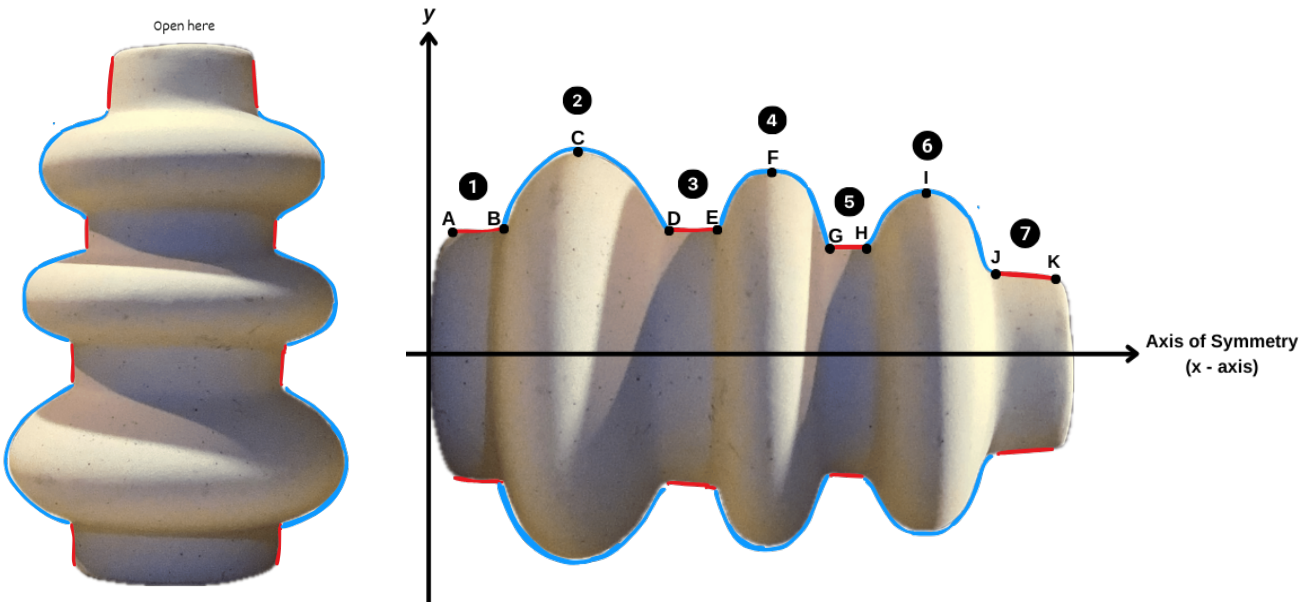

Figure 3: Idea of curves for finding volume of ceramic vase.

To determine the coordinates from (A to K); I used a simple and effective strategy, I will wind copper wire at different heights of the vase and measure the circumference (data in Appendix 2) to then give me the radius by dividing the circumference by 2π, (Figure 4). The radius becomes the measure of y coordinates and heights the x coordinates in centimetres, and height is the axis of symmetry.

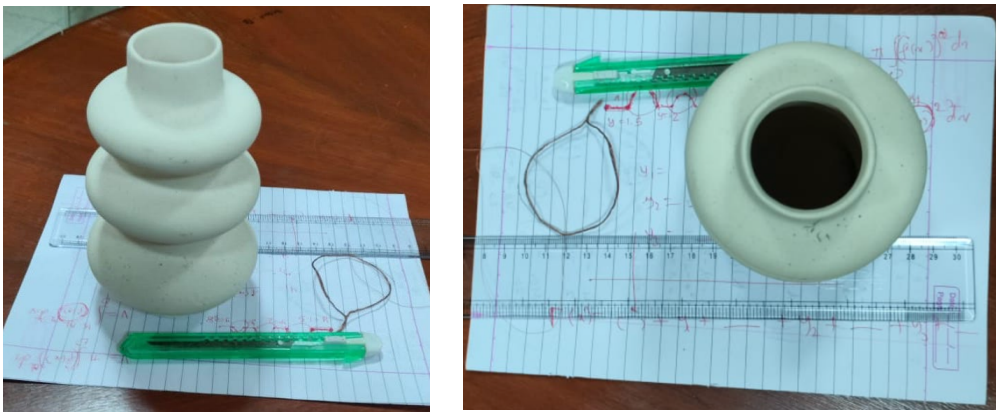

Figure 5: Method to find coordinates.

Coordinate	Co-ordinates x	Co-ordinates y
A	0	2.86
B	1.7	2.91
C	2.8	4.71
D	4.2	2.90
E	5.3	2.88
F	6.6	4.25
G	8.1	2.80
H	9.2	2.79
I	10.5	3.85
J	12.2	2.28
K	14.3	2.21

Table 1: Coordinates on the vase.

Finding the equations for the Parabolic curves (2, 4, 6).

Finding the equation for Curve 'BCD' using the formula of quadratic equations -

$y = ax^2 + bx + c$

Where –
a = concavity of parabola
b = $tilt\ of\ parabola$
c = $y-\ intercept\ of\ parabola$

$$y = ax^2 + bx + c$$

For point $B(1.7,2.91)$:
$$2.91 = a(1.7)^2 + b(1.7) + c \text{ .. (i)}$$

For point $C(2.8,4.71)$:
$$4.71 = a(2.8)^2 + b(2.8) + c \text{ .. (ii)}$$

For point $D(4.2,2.90)$:
$$2.90 = a(4.2)^2 + b(4.2) + c \text{ .. (iii)}$$

Note – That this is assuming that the curve follows a quadratic equation of the form –

$y = ax^2 + bx + c$

System of equations, Will be solved using substitution method to find the value of a, b, c.

$$2.91 = a(2.89) + 1.7b + c \text{ …... (i)}$$
$$4.71 = a(7.84) + 2.8b + c \text{ …...(ii)}$$
$$2.90 = a(17.64) + 4.2b + c \text{ …(iii)}$$

$$a = -1.1717, \quad b = 6.9090, \quad c = -5.4491$$

$$y = -1.1717x^2 + 6.9090x - 5.4491 \text{ ….. (Equation 2)}$$

Similarly Using the same method, I found the equations for curve 'EFG' and curve 'HIJ'

Equation for curve 'EFG' is, $y = -0.7216x^2 + 9.6410x - 27.9474$…… (Equation 4)

Equation for curve 'HIJ' is, $y = -0.5796x^2 + 12.2343x - 60.7046$…… (Equation 6)

Now I am going to find the equations for the vertical edges.

Finding the equation for Line 'AB' using the Point slope formula –

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

$$\textit{Substituting the values from points } A(0,2.86) \textit{ and } B(1.7,2.91)$$

$$y - 2.86 = \frac{2.91 - 2.86}{1.7 - 0}(x - 0)$$

$$y - 2.86 = \frac{0.05}{1.7}(x)$$

$$y - 2.86 = 0.0294x$$

$$y = 0.0294x + 2.86 \ldots \text{(Equation 1)}$$

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

Where –

$y - y_1$ = Vertical difference

$\frac{y_2 - y_1}{x_2 - x_1} = \textit{Slope of line}$

$(x - x_1) = \textit{Horizontal difference}$

Similarly, Using the point slope formula and substituting the values, I found the equations for the rest of the edges (1, 3, 5, 7).

Line DE: $y = -0.01818x + 3.676 \ldots (\textit{Equation } 3)$

Line GH: $y = -0.00909x + 2.8737 \ldots (\textit{Equation } 5)$

Line JK: $y = -0.03333x + 2.6867 \ldots (\textit{Equation } 7)$

Since, I have found 7 equations from the curves (2, 4, 6) and the edges (1, 3, 5, 7). I have arranged them in a piecewise function with their respective domains below.

$$f(x) = \begin{cases} 0.0294x + 2.86 & \text{for } 0 \le x \le 1.7 \\ -1.1717x^2 + 6.9090x - 5.4491 & \text{for } 1.7 \le x \le 4.2 \\ -0.01818x + 3.676 & \text{for } 4.2 \le x \le 5.3 \\ -0.7216x^2 + 9.6410x - 27.9474 & \text{for } 5.3 \le x \le 8.1 \\ -0.00909x + 2.8737 & \text{for } 8.1 \le x \le 9.2 \\ -0.5796x^2 + 12.2343x - 60.7046 & \text{for } 9.2 \le x \le 12.2 \\ -0.03333x + 2.6867 & \text{for } 12.2 \le x \le 14.3 \end{cases}$$

Finding the External Volume of the Vase using the formula of volume of revolution (Appendix 1), (Citation 2).

$$V = \pi \int_a^b [f(x)]^2 \, dx$$

Note –

$\int_a^b dx = \textit{definite integral}$

With ***a*** lower limit and ***b*** upper limit

$$V = \pi\left(\int_0^{1.7}(0.0294x + 2.86)^2\ dx\right) + \pi\left(\int_{1.7}^{4.2}(-1.1717x^2 + 6.9090x - 5.4491)^2\ dx\right)$$
$$+\ \pi\left(\int_{4.2}^{5.3}(-0.01818x + 3.676)^2\ dx\right) +\ \pi\left(\int_{5.3}^{8.1}(-0.7216x^2 + 9.6410x - 27.9474)^2\ dx\right)$$
$$+\ \pi\left(\int_{8.1}^{9.2}(-0.00909x + 2.8737)^2\ dx\right)$$
$$+\ \pi\left(\int_{9.2}^{12.2}(-0.5796x^2 + 12.2343x - 60.7046)^2\ dx\right)$$
$$+\ \pi\left(\int_{12.2}^{14.3}(-0.03333x + 2.6867)^2\ dx\right)$$

Let me divide the Integration in to 7 parts, $(I_1, I_2, I_3 \ldots)$

$$\boldsymbol{I_1} = \pi\int_0^{1.7}(0.0294x + 2.86)^2\ dx, = \pi\int_0^{1.7}(0.00086436x^2 + 0.167784x + 8.1796)\ dx,$$
$$= \pi\left[\frac{0.00086436x^3}{3} + \frac{0.167784x^2}{2} + 8.1796x\right]_0^{1.7} = \pi \cdot 14.150 = \boxed{44.45}\ (Using\ G.D.C)$$

$$\boldsymbol{I_2} = \pi\int_{1.7}^{4.2}(-1.1717x^2 + 6.9090x - 5.4491)^2\ dx, =$$
$$\pi\int_{1.7}^{4.2}(1.3723x^4 - 16.2003x^3 + 60.4858x^2 - 75.4878x + 29.6872)\ dx =$$
$$\pi\left[\frac{1.3723x^5}{5} - \frac{16.2003x^4}{4} + \frac{60.4858x^3}{3} - \frac{75.4878x^2}{2} + 29.6872x\right]_{1.7}^{4.2} = \pi \cdot 43.29 = \boxed{136.01}$$

Similarly, I found the values for $\boldsymbol{I_3}, \boldsymbol{I_4}, \boldsymbol{I_5}, \boldsymbol{I_6}, \boldsymbol{I_7}$

$$\boldsymbol{I_3} = \pi\int_{4.2}^{5.3}(-0.01818x + 3.676)^2\ dx = \boxed{46.31}\ (Using\ G.D.C)$$

$$\boldsymbol{I_4} = \pi\int_{5.3}^{8.1}(-0.7216x^2 + 9.6410x - 27.9474)^2\ dx, = \boxed{127.47}\ (Using\ G.D.C)$$

$$\boldsymbol{I_5} = \pi\int_{8.1}^{9.2}(-0.00909x + 2.8737)^2\ dx, = \boxed{26.99}\ (Using\ G.D.C)$$

$$\boldsymbol{I_6} = \pi\int_{9.2}^{12.2}(-0.5796x^2 + 12.2343x - 60.7046)^2\ dx, \quad = \boxed{111.17}\ (Using\ G.D.C)$$

$$\boldsymbol{I_7} = \pi\int_{12.2}^{14.3}(-0.03333x + 2.6867)^2\ dx = \boxed{33.25}\ (Using\ G.D.C)$$

$$\boldsymbol{V_{\text{total}} = I_1 + I_2 + I_3 + I_4 + I_5 + I_6 + I_7}$$

$$\boldsymbol{V_{\text{total}} = 44.45 + 136.01 + 46.31 + 127.47 + 26.99 + 111.17 + 33.25}$$

$$\boldsymbol{So, The\ total\ External\ volume\ is\ = \boxed{525.65\ cm^3}}$$

Finding the Internal Volume of the Ceramic Vase.

To find the internal volume, I decided to fill the vase with water, till the brim (Figure 6). And then poured the water carefully into a measuring cylinder (Figure 7) to figure out the Volume of the vase in milliliters and then converted it to cubic centimeters.

Figure 6: Water filled to the brim.

Figure 7: Water measurement using cylinder.

As seen the Internal Volume of the Vase is 400 ml, and when I convert it.

$$\mathbf{400\ mL = 400\ cm^3}$$

Finding the Surface Area of the Ceramic Vase.

Since I already, have the curves from the piecewise function above. I can find the Surface Area of the ceramic vase using the formula (Appendix 3), (Citation 3).

$$S = 2\pi \int_a^b f(x)\sqrt{1 + \left(\frac{df}{dx}\right)^2}\ dx$$

Note –

$\int_a^b dx = \ definite\ integral$

With ***a*** lower limit and ***b*** upper limit

$\left(\frac{df}{dx}\right)^2$ = derivative of function

First, I am going to find the **Surface area of the edges** using the formula above.

This is for Line 'AB'

$$f_1(x) = 0.0294x + 2.86 \quad \text{for } 0 \le x \le 1.7$$

$$f_1'(x) = \frac{d}{dx}(0.0294x + 2.86) = 0.0294$$

$Step\ 1: Calculate\ the\ Derivative\ (f_1'(x))$

$$S_1 = 2\pi \int_0^{1.7} f_1(x)\sqrt{1 + \left(f_1'(x)\right)^2}\ dx$$

$$S_1 = 2\pi \int_0^{1.7} (0.0294x + 2.86)\sqrt{1 + (0.0294)^2}\ dx$$

$Step\ 2: Substitute\ (f_1(x))\ and\ (f_1'(x))\ into\ the\ Integral$

$$Since\ (\sqrt{1 + (0.0294)^2})\ is\ a\ constant, we\ can\ approximate\ its\ value:$$

$$\sqrt{1 + (0.0294)^2} = \sqrt{1 + 0.00086436} = \sqrt{1.00086436} \approx 1.000432$$

$$Thus, the\ integral\ simplifies\ to:$$

$$S_1 \approx 2\pi \times 1.000432 \int_0^{1.7} (0.0294x + 2.86)\ dx$$

$$\int (0.0294x + 2.86)\ dx = \frac{0.0294}{2}x^2 + 2.86x + C$$

Step 3: Perform the Integration

Evaluate the definite integral from (0) to (1.7):

$$\left[\frac{0.0294}{2}(1.7)^2 + 2.86(1.7)\right] - [0 + 0] = \frac{0.0294}{2} \times 2.89 + 4.862 = 0.0426 + 4.862 = 4.9046$$

$$S_1 \approx 2\pi \times 1.000432 \times 4.9046 \approx 2\pi \times 4.9064 \approx \mathbf{30.830}\ \text{cm}^2$$

Using the same method, I found the surface areas for lines 'DE', 'GH', and 'JK' respectively.

Surface area for 'DE' or $y = -0.01818x + 3.676$ ⟶ $S_{\text{DE}} \approx 24.855\ \text{cm}^2$

Surface area for 'GH' or $y = -0.00909x + 2.8737$ ⟶ $S_{\text{GH}} \approx 19.191\ \text{cm}^2$

Surface area for 'JK' or $y = -0.03333x + 2.6867$ ⟶ $S_{\text{JK}} \approx 30.580\ \text{cm}^2$

Now I am going to find the **Surface area of the curves** using the same formula above.

This is for curve 'BCD'

$$f_2(x) = -1.1717x^2 + 6.9090x - 5.4491 \quad \text{for } 1.7 \le x \le 4.2$$

Step 1: Calculate the Derivative ($f_1'(x)$)

$$f_2'(x) = \frac{d}{dx}(-1.1717x^2 + 6.9090x - 5.4491) = -2.3434x + 6.9090$$

Step 2: Substitute ($f_2(x)$) and ($f_2'(x)$) into the Integral

$$S_2 = 2\pi \int_{1.7}^{4.2} f_2(x)\sqrt{1 + \left(f_2'(x)\right)^2}\ dx$$

$$S_2 = 2\pi \int_{1.7}^{4.2} (-1.1717x^2 + 6.9090x - 5.4491)\sqrt{1 + (-2.3434x + 6.9090)^2}\ dx$$

$$\sqrt{1 + (-2.3434x + 6.9090)^2} = \sqrt{1 + (-2.3434x + 6.9090)^2}$$

$$= \sqrt{1 + (5.4825x^2 - 32.3172x + 47.4783)} = \sqrt{5.4825x^2 - 32.3172x + 48.4783}$$

$$S_2 = 2\pi \int_{1.7}^{4.2} (-1.1717x^2 + 6.9090x - 5.4491)\sqrt{5.4825x^2 - 32.3172x + 48.4783}\ dx$$

Given the complexity of the integral, I will have to apply the Trapezoidal Rule (Citation 4) with N = 2, in order to manually calculate the surface area of the curve 'BCD'.

$$a = 1.7, \quad b = 4.2, \quad n = 2, \quad h = \frac{4.2 - 1.7}{2} = 1.25$$

Trapezoidal Rule Formula:

$$\int_a^b f(x)\ dx \approx \frac{h}{2}[f(a) + 2f(a + h) + f(b)]$$

$$f(x) = (-1.1717x^2 + 6.9090x - 5.4491)\sqrt{5.4825x^2 - 32.3172x + 48.4783}$$

$\boxed{For\ f(a)}$

$$f(1.7) = (-1.1717(1.7)^2 + 6.9090(1.7) - 5.4491)\sqrt{5.4825(1.7)^2 - 32.3172(1.7) + 48.4783}$$
$$= (-1.1717 \times 2.89 + 11.7553 - 5.4491)\sqrt{5.4825 \times 2.89 - 32.3172 \times 1.7 + 48.4783}$$
$$= (-3.3927 + 11.7553 - 5.4491)\sqrt{15.8183 - 54.5472 + 48.4783}$$

$$= 2.9135 \times \sqrt{15.8183 - 54.5472 + 48.4783} = 2.9135 \times \sqrt{9.7494} \approx 2.9135 \times 3.1233 \approx 9.097\ \text{cm}^2$$

$\boxed{\begin{matrix} For \\ f(a+h) \end{matrix}}$

$$f(2.95) = (-1.1717(2.95)^2 + 6.9090(2.95) - 5.4491)\sqrt{5.4825(2.95)^2 - 32.3172(2.95) + 48.4783}$$
$$= (-1.1717 \times 8.7025 + 20.3735 - 5.4491)\sqrt{5.4825 \times 8.7025 - 32.3172 \times 2.95 + 48.4783}$$
$$= (-10.1902 + 20.3735 - 5.4491)\sqrt{47.6533 - 95.3141 + 48.4783}$$

$$= 4.7342 \times \sqrt{0.8175} \approx 4.7342 \times 0.9041 \approx 4.281\ \text{cm}^2$$

$\boxed{For\ f(b)}$

$$f(4.2) = (-1.1717(4.2)^2 + 6.9090(4.2) - 5.4491)\sqrt{5.4825(4.2)^2 - 32.3172(4.2) + 48.4783}$$
$$= (-1.1717 \times 17.64 + 29.0598 - 5.4491)\sqrt{5.4825 \times 17.64 - 32.3172 \times 4.2 + 48.4783}$$
$$= (-20.6651 + 29.0598 - 5.4491)\sqrt{96.5871 - 135.3912 + 48.4783}$$

$$= 2.9456 \times \sqrt{9.6742} \approx 2.9456 \times 3.1136 \approx 9.179\ \text{cm}^2$$

$$\int_{1.7}^{4.2} f(x)\ dx \approx \frac{1.25}{2}[f(1.7) + 2f(2.95) + f(4.2)] = \frac{1.25}{2}[9.097 + 2 \times 4.281 + 9.179]$$

$$= 0.625 \times [9.097 + 8.562 + 9.179] = 0.625 \times 26.838 \approx 16.774\ \text{cm}^2$$

$$S_{\text{BCD}} \approx 2\pi \times 16.774 \approx 105.376\ \text{cm}^2$$

In order to verify this answer, I used wolfram math software (Citation 7) to solve the Integral.

Computation result

$$2\pi \int_{1.7}^{4.2} (-1.1717x^2 + 6.9090x - 5.4491)\sqrt{5.4825x^2 - 32.3172x + 48.4783}\ dx =$$
111.483

As seen, the actual answer is 111.48 square cm. My answer is not very accurate and to improve the accuracy I will have to increase the value of N.

Using the method, I found the surface area integrals for curves 'EFG', and 'HIJ' respectively, and then obtained the final answer using Wolfram Math.

Surface area for 'EFG' or $y = -0.7216x^2 + 9.6410x - 27.9474 \longrightarrow$ $S_{\text{EFG}} \approx 96.09\ \text{cm}^2$

Surface area for 'HIJ' or $y = -0.5796x^2 + 12.2343x - 60.7046 \longrightarrow$ $S_{\text{HIJ}} \approx 72.50\ \text{cm}^2$

Hence, the Total Surface Area for the Vase is equal to.

$$S_{\text{total}} = S_{\text{AB}} + S_{\text{BCD}} + S_{\text{DE}} + S_{\text{EFG}} + S_{\text{GH}} + S_{\text{HIJ}} + S_{\text{JK}}$$
$$\approx 30.83 + 111.48 + 24.86 + 96.09 + 19.19 + 72.5 + 30.58$$

$$= \mathbf{385.53\ cm^2}$$

Finding the Cost of ceramic used to make the Vase.

After some research, (Citation 5) I found that it is called a "White Wave Vase". And the type of ceramic used is called "Feldspar" which has a density of around $\rho = 3.2\ \mathrm{g/cm^3}$, and a cost of around 40 rupees per Kg.

First, I need to find the volume of the feldspar ceramic used in to make the vase, for that I found the external and internal volume of the vase. So, volume of ceramic is.

$$V_{\text{ceramic}} = V_{\text{external}} - V_{\text{internal}} = 523.89\ \mathrm{cm^3} - 400\ \mathrm{cm^3} = 123.89\ \mathrm{cm^3}$$

Finding the mass using density formula

$$m = V \times \rho = 123.89\ \mathrm{cm^3} \times 3.2\ \mathrm{g/cm^3} = \mathbf{374.048\ g}$$

The website has given the mass to be **400g**.

Converting from grams to Kg's.

$$m = \frac{374.048\ \mathrm{g}}{1000} = 0.374048\ \mathrm{kg}$$

$$\mathrm{Cost}_{ceramic} = 0.374048\ \mathrm{kg} \times 40\ \mathrm{rupees/kg}$$

$$= \mathbf{14.9619\ rupees}$$

$$\text{Percentage Error} = \left(\frac{|\text{Experimental Value} - \text{Theoretical Value}|}{|\text{Theoretical Value}|}\right) \times 100\%$$

(Citation 9)

$$\mathrm{E} = 374.04\ \mathrm{g}, \quad \mathrm{T} = 400\ \mathrm{g}$$

$$\text{Absolute Error} = |\mathrm{E} - \mathrm{T}| = |374.04\ \mathrm{g} - 400\ \mathrm{g}| = 25.96\ \mathrm{g}$$

$$\text{Percentage Error} = \left(\frac{25.96\ g}{400\ g}\right) \times 100\% = 6.49\%$$

Finding the Number of Vases made in 1Kg of ceramic material.

My uncle told me that, in their industry approximately 20% of the ceramic material goes to waste, while making a vase, so in my calculations I need to take this into consideration as well.

$$m_{\text{ceramic}} = 0.374048\ \mathrm{kg/vase}$$

Waste percentage.

$$W = 20\% = 0.20$$

$$\text{Effective Ceramic} = 1\ \mathrm{kg} \times (1 - W) = 1\ \mathrm{kg} \times (1 - 0.20) = 0.80\ \mathrm{kg}$$

$$\text{Number of Vases per kg} = \frac{\text{Effective Ceramic}}{m_{\text{ceramic}}} = \frac{0.80\ \mathrm{kg}}{0.374048\ \mathrm{kg/vase}} \approx 2.14\ \mathrm{vases/kg}$$

Considering a 20% material waste, approximately **2 vases** can be produced per kilogram of feldspar ceramic.

So, the actual cost per-vase is =

$$C_t = \frac{C_k}{N} = \frac{40\ \mathrm{rupees/kg}}{2\ \mathrm{vases/kg}} = 20\ \mathrm{rupees/vase}$$

Note –

$C_r = 15$ rupees (Cost of ceramic per vase)

$N = 2$ vases/kg (Number of vases produced per kilogram)

$C_k = 40$ rupees/kg (Cost per kilogram of ceramic)

So, then I found that -

$$C_o = C_t - C_r = 20\ \mathrm{rupees/vase} - 15\ \mathrm{rupees/vase} = 5\ \mathrm{rupees/vase}$$

the extra 5 rupees per vase, is going to waste. And this will be kept into consideration while optimizing the vase.

Optimizing the Vase based on its functions.

In order to optimize the vase, to reduce the cost and improve the efficiency in terms of the number of vases made per kg of feldspar ceramic.

I will set an optimization goal, I will try to increase the number of vases per kg, from 2 to 3. By optimizing the surface area which will reduce material usage. I will also try to maintain the volume of the vase and try to keep a similar aesthetic appeal for the vase.

To achieve this goal, I will use the **calculus-based optimization** (citation 6), method to derive a new piecewise function, for which I will compare the volume, surface area, and the cost of the ceramic for the new vase. (I will also plot the new vase).

I already found the piecewise function for the original vase, which is below.

$$f(x) = \begin{cases} 0.0294x + 2.86 & \text{for } 0 \le x \le 1.7 \\ -1.1717x^2 + 6.9090x - 5.4491 & \text{for } 1.7 \le x \le 4.2 \\ -0.01818x + 3.676 & \text{for } 4.2 \le x \le 5.3 \\ -0.7216x^2 + 9.6410x - 27.9474 & \text{for } 5.3 \le x \le 8.1 \\ -0.00909x + 2.8737 & \text{for } 8.1 \le x \le 9.2 \\ -0.5796x^2 + 12.2343x - 60.7046 & \text{for } 9.2 \le x \le 12.2 \\ -0.03333x + 2.6867 & \text{for } 12.2 \le x \le 14.3 \end{cases}$$

I am going to Optimize segment AB, using calculus-based optimization.

Note: Ensuring derivative continuity may require significant adjustments. For material optimization, So I may prioritize function value continuity and allow a slight derivative discontinuity.

$$f_{\text{AB}}(x) = 0.0294x + 2.86 \quad \text{for} \quad 0 \le x \le 1.7$$

$$\text{Function Value Continuity:} \quad f_{\text{AB}}(1.7) = f_{\text{BCD}}(1.7)$$

$$\text{Derivative Continuity:} \quad f'_{\text{AB}}(1.7) = f'_{\text{BCD}}(1.7)$$

Calculating the original function value at x = 1.7

$$f_{\text{AB}}(1.7) = 0.0294 \times 1.7 + 2.86 = 0.0500 + 2.86 = 2.91$$

$$m_{\text{opt}} = 0.0277$$

To reduce material usage, I will have to flatten the slope of Segment AB. Let's assume a new slope less than the original.

$$f_{\text{AB, optimized}}(1.7) = m_{\text{opt}} \times 1.7 + c_{\text{opt}} = 2.74$$

Using **function value continuity**, solve for the new intercept

$$0.0277 \times 1.7 + c_{\text{opt}} = 2.74$$

$$0.04709 + c_{\text{opt}} = 2.74$$

$$c_{\text{opt}} = 2.74 - 0.04709 = 2.69291 \approx 2.700$$

$$\boldsymbol{f_{\text{AB, optimized}}(x) = 0.0277x + 2.700} \quad \text{for} \quad 0 \le x \le 1.7$$

Final AB optimized Eqn.

$$f_{\text{AB, optimized}}(1.7) = 0.0277 \times 1.7 + 2.700 = 0.04709 + 2.700 = 2.74709 \approx 2.747$$

$$f_{\text{BCD, optimized}}(1.7) = 2.7586$$

Assuming $f_{\text{BCD, optimized}}(1.7) = 2.7586$,

the difference is negligible (≈ 0.0116),

which is acceptable for optimization.

Now using the same method, I am going to optimize the curve BCD.

$$f_{\text{BCD}}(x) = -1.1717x^2 + 6.0909x - 5.4491 \quad \text{for} \quad 1.7 \leq x \leq 4.2$$

To ensure a **smooth transition** between Parabolic Curve BCD and its adjacent segments (AB and DE), the equation must satisfy the following at the **transition points**:

$$\textit{Function Value Continuity at } (x = 1.7):$$

$$f_{\text{BCD}}(1.7) = f_{\text{AB, optimized}}(1.7)$$

$$\textit{Derivative Continuity at } (x = 1.7):$$

$$f'_{\text{BCD}}(1.7) = f'_{\text{AB, optimized}}(1.7)$$

$$\textit{Function Value Continuity at } (x = 4.2):$$

$$f_{\text{BCD}}(4.2) = f_{\text{DE, optimized}}(4.2)$$

$$\textit{Derivative Continuity at } (x = 4.2):$$

$$f'_{\text{BCD}}(4.2) = f'_{\text{DE, optimized}}(4.2)$$

Note: Ensuring derivative continuity may require significant adjustments. For material optimization, So I may prioritize function value continuity and allow a slight derivative discontinuity.

$\boldsymbol{Function\ Value\ at\ Transition\ Points\ At(x = 1.7)}$:

$f_{\text{AB, optimized}}(1.7) = 0.0277 \times 1.7 + 2.700 = 0.04709 + 2.700 = 2.74709 \approx 2.747$

$f_{\text{BCD, optimized}}(1.7) = 2.7586$

Difference $= 2.7586 - 2.747 \approx 0.0116$ (Negligible and acceptable)

Note: For the purpose of this derivation, I'll assume the optimized Segment DE maintains continuity with the optimized Segment BCD. If specific optimized coefficients for DE are provided, they should be used here.

To **reduce material usage**, I will **flatten** the curvature of Parabolic Curve BCD by **reducing the absolute value of the quadratic coefficient**. Let's assume a **new quadratic coefficient** less steep than the original.

$$a_{\text{opt}} = -1.1047$$

$$f_{\text{BCD, optimized}}(x) = -1.1047x^2 + b_{\text{opt}}x + c_{\text{opt}} \quad \text{for} \quad 1.7 \leq x \leq 4.2$$

Now I will find b_{opt} *and* c_{opt} -

$\boldsymbol{Apply\ Function\ Value\ Continuity\ at\ (x = 1.7)}$

$$f_{\text{BCD, optimized}}(1.7) = f_{\text{AB, optimized}}(1.7) = 2.7586$$

$$-1.1047 \times (1.7)^2 + b_{\text{opt}} \times 1.7 + c_{\text{opt}} = 2.7586$$

$$-1.1047 \times 2.89 + 1.7b_{\text{opt}} + c_{\text{opt}} = 2.7586$$

$$-3.1932 + 1.7b_{\text{opt}} + c_{\text{opt}} = 2.7586 \quad \text{(Equation 1)}$$

Apply Derivative Continuity at ($\boldsymbol{x = 1.7}$)

$$f'_{\text{BCD, optimized}}(x) = \frac{d}{dx}\left(-1.1047x^2 + b_{\text{opt}}x + c_{\text{opt}}\right) = -2.2094x + b_{\text{opt}}$$

$$f'_{\text{BCD, optimized}}(1.7) = -2.2094 \times 1.7 + b_{\text{opt}} = 2.7610$$

$$-3.7560 + b_{\text{opt}} = 2.7610$$

$$b_{\text{opt}} = 2.7610 + 3.7560 = 6.5170 \quad \text{(Equation 2)}$$

Solve for $(\boldsymbol{b_{\text{opt}}})$ ***and*** $(\boldsymbol{c_{\text{opt}}})$

$$\textit{Equation } 2\text{: } b_{\text{opt}} = 6.5170$$

$$-3.1932 + 1.7 \times 6.5170 + c_{\text{opt}} = 2.7586$$

$$-3.1932 + 11.0890 + c_{\text{opt}} = 2.7586$$

$$7.8958 + c_{\text{opt}} = 2.7586$$

$$c_{\text{opt}} = 2.7586 - 7.8958 = -5.1372$$

Substitute (b_{opt}) *into Equation* 1:

Formulate the Optimized Parabolic Curve BCD Equation

$$f_{\text{BCD, optimized}}(x) = -1.1047x^2 + 6.5170x - 5.1372 \quad \text{for} \quad 1.7 \le x \le 4.2$$

$$f_{\text{BCD, optimized}}(x) = \boldsymbol{-1.1047x^2 + 6.517x - 5.136} \quad \textbf{for} \quad \boldsymbol{1.7 \le x \le 4.2}$$

Rounded of equation

Using the Same method, I found the optimized equations for the rest of the edges and curves, and then I complied a full optimized piecewise function.

$$f_{\text{optimized}}(x) = \begin{cases} 0.0277x + 2.700 & \text{for} \quad 0 \le x \le 1.7 \\ -1.1047x^2 + 6.517x - 5.136 & \text{for} \quad 1.7 \le x \le 4.2 \\ 0.5995x - 0.1155 & \text{for} \quad 4.2 \le x \le 5.3 \\ -0.6810x^2 + 9.1038x - 26.3896 & \text{for} \quad 5.3 \le x \le 8.1 \\ -0.0086x + 2.7177 & \text{for} \quad 8.1 \le x \le 9.2 \\ -0.5469x^2 + 11.5314x - 57.2259 & \text{for} \quad 9.2 \le x \le 12.2 \\ -0.0313x + 2.542 & \text{for} \quad 12.2 \le x \le 14.3 \end{cases}$$

Now I am going to **plot** the optimized functions above to visualize the final vase using GeoGebra (Citation 7) software, to ensure aesthetic appeal and continuity in the optimized functions.

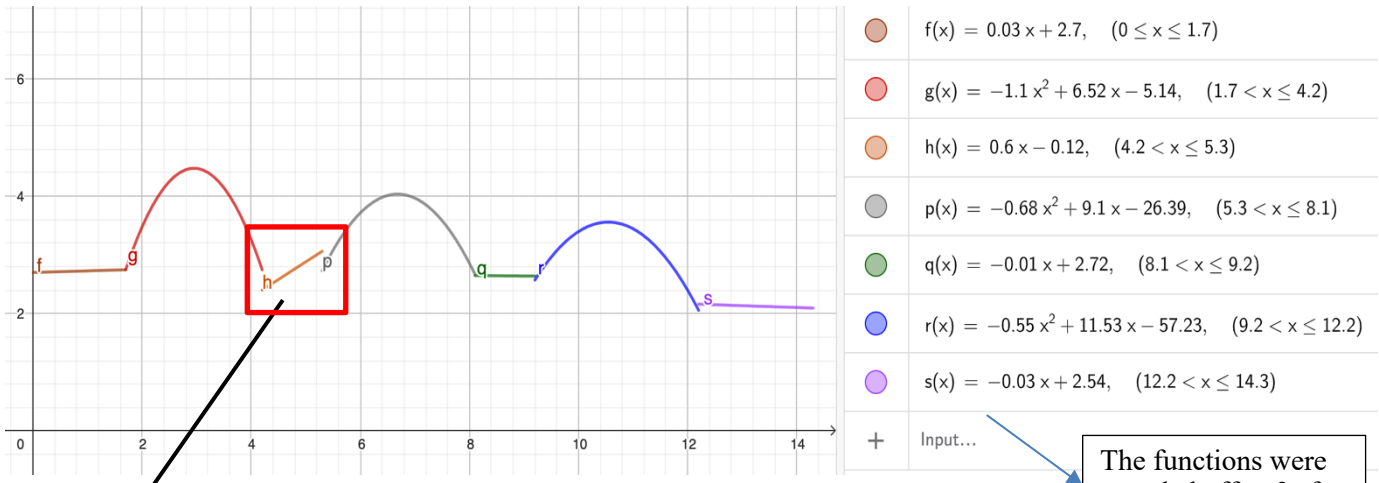

Figure 8: Optimized Vase Plot

The functions were rounded off to 3s.f

I have made an error while trying to optimize equation 3, h(x). I was not able to accurately ensure smooth continuity, due to the assumptions (which was the difference in the slope of original and optimized equations) taken while performing the calculus-based optimization.

In-order to remove this breakage, and ensure smooth continuity, I will construct a line segment AB, that connects both quadratic equations.

Line segment AB has been added (as shown in the figure), to ensure continuity in the design of the vase.

The Eqn of Line segment AB is –

$$\boldsymbol{f(x) = 0.01x + 2.72},\ 4.2 \leq x \leq 5.31$$

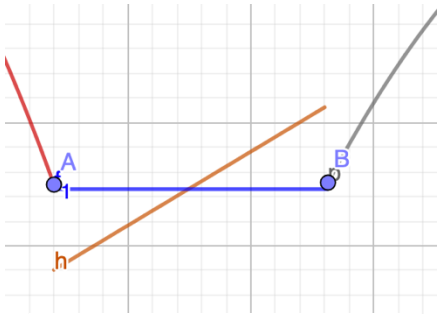

Figure 9: Line Segment AB

Now I will delete the function h(x) and replace it with t(x).

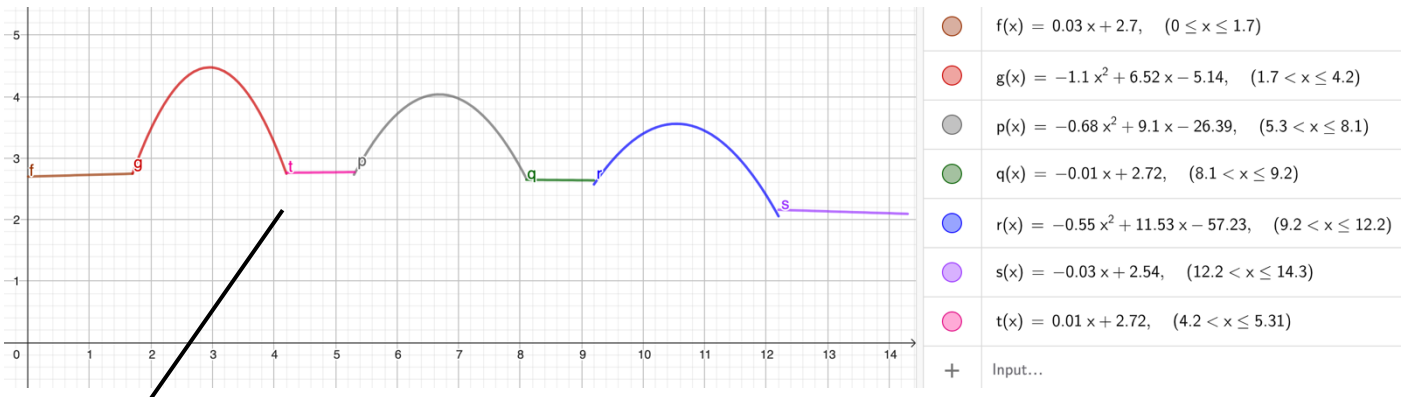

Figure 10: Corrected Vase Design

Now that the design is correct and a smooth continuity is ensured throughout the vase, I will procced with, building **3D designs** of the new optimized vase.

To better visualize the optimized vase, I have built 3D models of the vase by rotating the functions around the x axis, by a factor of 2π.

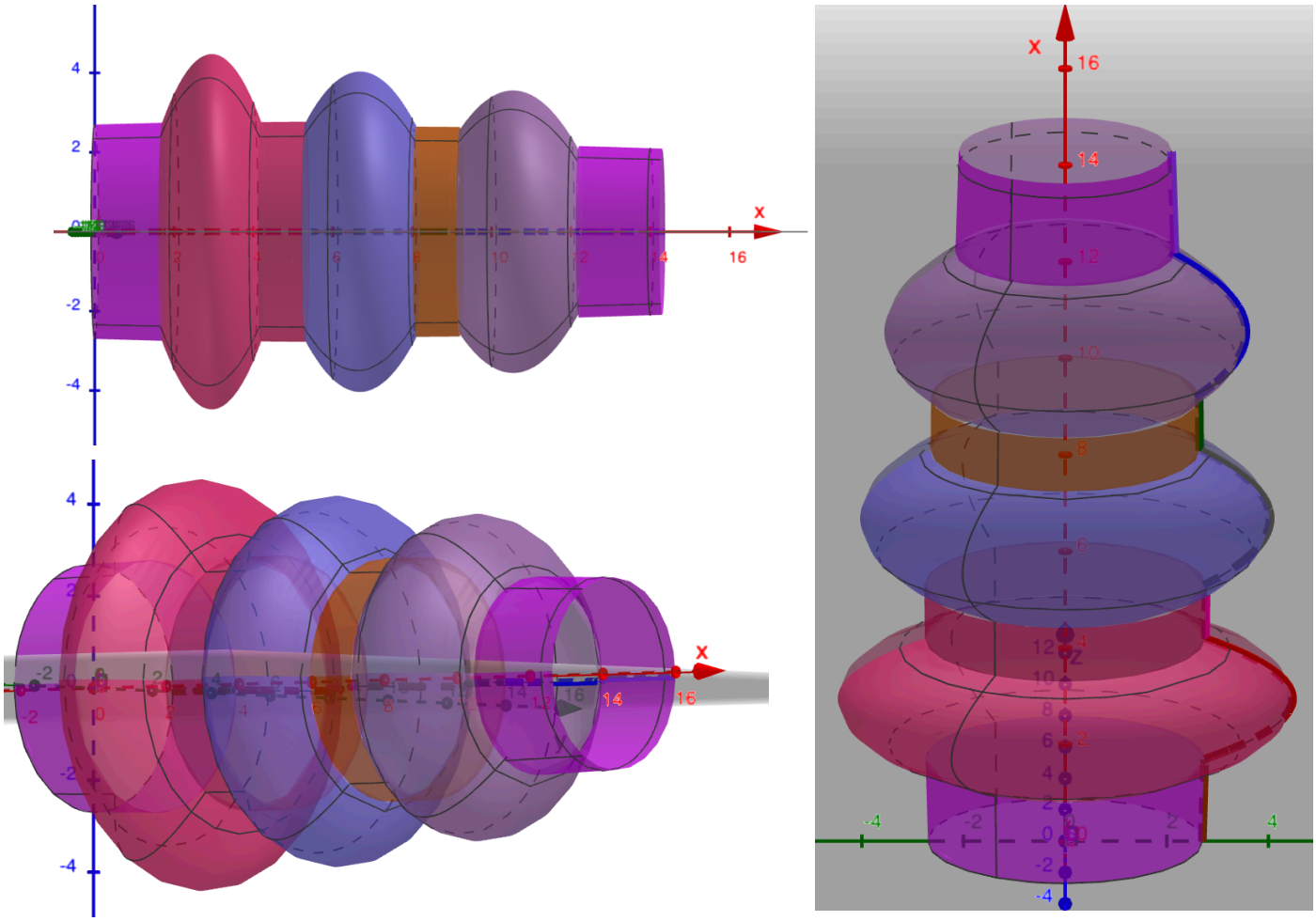

Figure 10: 3D Vase Design

As shown in the designs above, I have kept the same design during optimization, as it is easier for my uncle's company to produce the same design, I just need to make sure that the overall cost per vase decreases and if the company can produce 3 vases in 1Kg of feldspar ceramic, by reducing the volume and surface area in the same aesthetic appeal to the customer.

Now I am going to compare the volume, surface area and the cost of ceramic for this new and optimized vase.

I have complied a new piecewise function with the correction of Line Segment AB in the third eqn.

$$f_{\text{optimized}}(x) = \begin{cases} 0.0277x + 2.700 & \text{for } 0 \leq x \leq 1.7 \\ -1.1047x^2 + 6.517x - 5.136 & \text{for } 1.7 < x \leq 4.2 \\ 0.01x + 2.72 & \text{for } 4.2 \leq x \leq 5.31 \\ -0.6810x^2 + 9.1038x - 26.3896 & \text{for } 5.3 < x \leq 8.1 \\ -0.0086x + 2.7177 & \text{for } 8.1 < x \leq 9.2 \\ -0.5469x^2 + 11.5314x - 57.2259 & \text{for } 9.2 < x \leq 12.2 \\ -0.0313x + 2.542 & \text{for } 12.2 < x \leq 14.3 \end{cases}$$

Now to find the **Volume and Surface Area of this new optimized vase**.

In order to solve it faster and to reduce manual calculation, I wrote a python script to find surface area and volume (Appendix 4), which gives me $V \approx \mathbf{450.6\ cm^3}$, and $S \approx \mathbf{365.48\ cm^3}$. However, to verify the accuracy of the script I used wolfram math to solve the equations.

To find the **volume** I have used the same method as I used before (volume of revolution).

$$V = \pi \int_a^b [f(x)]^2\ dx$$

$$V = \pi\left(\int_0^{1.7} (0.0277x + 2.700)^2 dx + \int_{1.7}^{4.2} (-1.1047x^2 + 6.517x - 5.136)^2 dx + \int_{4.2}^{5.3} (0.01x - 2.72)^2 dx + \int_{5.3}^{8.1} (-0.6810x^2 + 9.1038x - 26.3896)^2 dx + \int_{8.1}^{9.2} (-0.0086x + 2.7177)^2 dx + \int_{9.2}^{12.2} (-0.5469x^2 + 11.5314x - 57.2259)^2 dx + \int_{12.2}^{14.3} (-0.0313x + 2.542)^2 dx\right)$$

I verified this answer using Wolfram Math (citation 7), I found the value of each integral without multiplying π and then added them to find the Sum.

$$\text{Sum} = 12.61 + 38.68 + 8.25 + 36.50 + 7.60 + 29.96 + 9.50 = 143.10$$

$$V = \pi \times 143.10 \approx \mathbf{450.82\ cm^3}$$

To find the **Surface area** I have used the same method as I used before (Surface area of revolution).

$$S = 2\pi \int_a^b f(x) \sqrt{1 + \left(\frac{df}{dx}\right)^2}\ dx$$

Then I found the integral for each function by applying it to the surface area formula.

$$S = 2\pi\left(\int_0^{1.7} (0.0277x + 2.700)\sqrt{1 + (0.0277)^2}\ dx + \int_{1.7}^{4.2} (-1.1047x^2 + 6.517x - 5.136)\sqrt{1 + (-2.2094x + 6.517)^2}\ dx + \int_{4.2}^{5.3} (0.01x + 2.72)\sqrt{1 + (0.01)^2}\ dx + \int_{5.3}^{8.1} (-0.6810x^2 + 9.1038x - 26.3896)\sqrt{1 + (-1.362x + 9.1038)^2}\ dx + \int_{8.1}^{9.2} (-0.0086x + 2.7177)\sqrt{1 + (-0.0086)^2}\ dx + \int_{9.2}^{12.2} (-0.5469x^2 + 11.5314x - 57.2259)\sqrt{1 + (-1.0938x + 11.5314)^2}\ dx$$

$$+\int_{12.2}^{14.3}(-0.0313x+2.542)\sqrt{1+(-0.0313)^2}\,dx\Bigg)$$

Using Wolfram Math (Citation 8), I found the value of each integral (with 2 π) and then added them to find the Sum.

$$\text{Sum} = 29.10 + 104.46 + 20.01 + 88.74 + 18.26 + 77.89 + 28.08$$

$$= \mathbf{365.54\ cm^3}$$

Now I am going to find the **cost of ceramic**, and **number of vases produced** in 1Kg of feldspar ceramic.

I already know the cost of the ceramic is around **40 rupees per kg**, and density is $\boldsymbol{\rho = 3.2\ g/cm^3}$, this is constant for both the original and optimized vase.

Original Vase specifications –

- $V_{\text{external}} = 523.89\ \text{cm}^3$
- $V_{\text{internal}} = 400\ \text{cm}^3$
- $S_{area} = 385.53\ \text{cm}^2$
- $V_{\text{ceramic}} = 123.89\ \text{cm}^3$
- $m_{\text{ceramic}} = V \times \rho = 374.048\ \text{g}$
- $C_r = 20$ rupees (Cost of ceramic per vase)
- $N = 2$ vases/kg (Number of vases produced per kilogram)

Note –

The goal for optimization, was **3 vases** per kg of feldspar ceramic.

My uncle, found a way to reuse the waste ceramic in the production of the other vase in the same batch. So hence the wastage is negligible for this reason I am going to assume, that there is **0% waste**.

I know that the **optimized vase** has a Volume and Surface area of $V \approx \mathbf{450.8\ cm^3}$ and $S \approx \mathbf{365.54\ cm^3}$

To find the volume of ceramic used in the new vase, $V_{\text{ceramic2}} = V_{\text{external2}} - V_{\text{internal2}}$, Since $V_{\text{internal2}}$ is unknown, I will derive it based on the reduced ceramic volume required to meet the production goal (3 vases per kg).

Hence, the desired mass of ceramic per vase is:

$$m_{\text{desired}} = \frac{1000\ \text{g}}{3} \approx 333.33\ \text{g/vase}$$

So then, the required ceramic volume per vase is:

$$V_{\text{ceramic2}} = \frac{m_{\text{desired}}}{\rho} = \frac{333.33\ \text{g}}{3.2\ \text{g/cm}^3} \approx 104.17\ \text{cm}^3$$

So now the mass of ceramic per optimized vase is:

$$m_{\text{ceramic2}} = V_{\text{ceramic2}} \times \rho = 104.17\ \text{cm}^3 \times 3.2\ \text{g/cm}^3 = 333.34\ \text{g}$$

Then the cost per vase is:

$$\text{Cost}_{\text{optimized}} = \frac{333.34\ \text{g}}{1000\ \text{g/kg}} \times ₹40/\text{kg} = \mathbf{₹13.33}$$

And number of vases produces is:

$$N_{optimized} = \frac{1000\ \text{g}}{333.34\ \text{g/vase}} \approx \mathbf{3\ vases/kg}$$

Hence, I have **met the goal** of optimization of **3 vases per kg**, I have also reduced the cost per vase,

From **20 rupees (including waste), to 13.33 rupees (negligible waste).**

For a better understanding of the result I have compiled a table -

Parameter	Original Vase	Optimized Vase
External Volume (cm^3)	523.89	450.82
Internal Volume (cm^3)	400	332.66
Surface Area (cm^2)	385.53	365.54
Ceramic Volume (cm^3)	123.89	104.17
Mass of Ceramic (g)	396.45	333.34
Cost per Vase (₹)	20.00	13.33
Vases per kg	2.52	3

Strengths, Limitations & Conclusion -

Strengths –

One of the most important strengths of this exploration is that I learned how to efficiently optimize and design/model solutions based of real-world problems through effective use of mathematical tools like (functions, integration & differentiation) and technology like (GeoGebra, and python) to solve such issues. I was also able to find coordinates based on the height as x-axis, and radius as y-axis, to model the vase. Moreover, the procedure of optimizing a vase using functions and calculus, is very simple to replicate and understand and can be used in other companies like my uncles who produce such decorative items, it can also be extended to other parts of material sciences and cost optimization based on the industry.

Limitations –

Overall, there were a lot of limitations to perform this exploration, measuring the vase height and radius to determine the coordinates was very hard to execute, it also resulted in many human errors as measuring with the copper wire was not very accurate, I had to repeat the process 4 to 5 times to cross check and verify the data obtained and figure out the exact values. Moreover, while calculating the cost and the weight of the ceramic material used, I had to assume that there were no impurities or any sort of mixture or combination of materials, this is a big issue when it comes to the real-world solutions. So hence I had to assume a fixed density constant for that one type of material (feldspar ceramic). It was also very difficult to take into consideration the exact waste percentage; my uncle was only able to give me an approximate idea, I have also not taken into consideration the thickness of the vase, and how it would affect the quality and durability of the product produced.

Conclusion –

I wanted to optimize a ceramic vase, produced by my uncle's company to help him reduce cost and to produce at least 3 vases per kilogram of ceramic. I achieved the objective by using calculus-based optimization to reduce the volume and surface area of the vase, while maintaining the same visual and aesthetic appeal for the ease of production. The most fascinating part about working on this exploration was that I found a solution to a real-world problem faced by my uncle's company using purely the skill of my mathematical comprehension. The exploration allowed me to explore various aspects like practically working with finding volume and surface area of functions, working with definite integration, trapezoidal rule and calculus-based optimization. The IA also taught me how to use different mathematical tools, like GeoGebra, Wolfram-Math and mathematical libraries in python, it also taught me how to properly present a problem and design a solution. In the exploration, I have used various illustrations and visuals, including notes and variables to make the solution more comprehensible. Moreover, all calculations, figures, code and tables have been added. Overall, I feel that the IA has been an opportunity that I learned from, to design and tackle real-world problems like this one, I wish to look into more problems and work on their mathematical solutions.

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Appendix 1

Volume of Revolution about y-axis

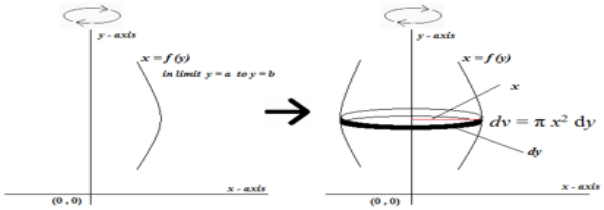


Figure : Volume Measurement idea.

The volume of the disc element is $dv = \pi\, x^2\, dy$

Therefore the complete volume V is given as $\int_{y=a}^{y=b} dv = \int_{y=a}^{y=b} \pi\, x^2 dy$ where $x = f(y)$

Appendix 2

This is the data for the Circumference and its corresponding Heights,

Height	Circumference
0.0	18.0
1.7	18.3
2.1	29.6
4.2	18.2
5.3	18.1
6.6	26.7
8.1	17.6
9.2	17.5
10.5	24.2
12.2	14.4
14.3	13.9

Appendix 3

Surface Area Formula

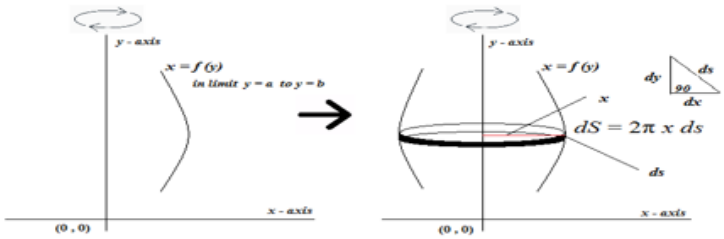


Figure : Surface Area measurement idea.

The surface area element dS = $2\pi\, x\, ds$. Where

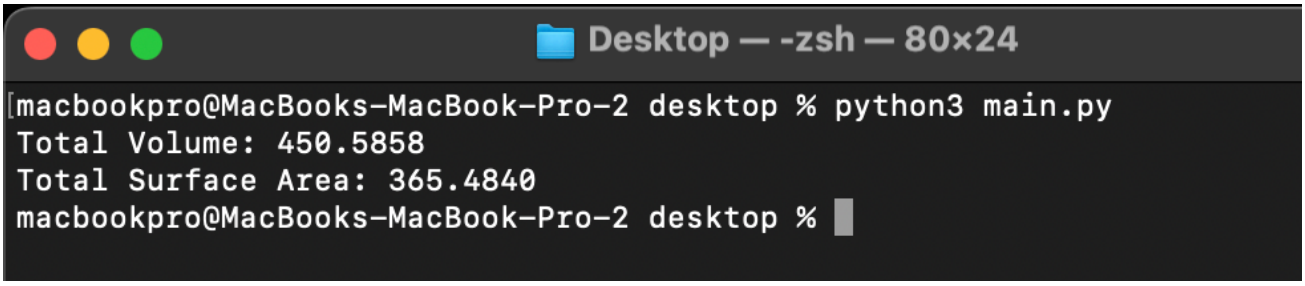
$$ds^2 = dx^2 + dy^2 \quad \rightarrow \quad \frac{ds}{dx} = \sqrt{1 + \left(\frac{dx}{dy}\right)^2}.$$

Therefore the complete surface area is $S = 2\pi \int_{y=a}^{y=b} x \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\; dx.$

Appendix 4 – Volume and surface area code,

```
import numpy as np
from scipy.integrate import quad

# Piecewise function for radius
def f(x):
    if x <= 1.7: return 0.0277 * x + 2.700
    if x <= 4.2: return -1.1047 * x**2 + 6.517 * x - 5.136
    if x <= 5.31: return 0.01 * x + 2.72
    if x <= 8.1: return -0.6810 * x**2 + 9.1038 * x - 26.3896
    if x <= 9.2: return -0.0086 * x + 2.7177
    if x <= 12.2: return -0.5469 * x**2 + 11.5314 * x - 57.2259
    return -0.0313 * x + 2.542

# Numerical derivative using central difference method
def df(x, h=1e-5):
    return (f(x + h) - f(x - h)) / (2 * h)

# Volume: V = π ∫ f(x)^2 dx
def volume(x1, x2):
    return quad(lambda x: np.pi * f(x)**2, x1, x2)[0]

# Surface area: A = 2π ∫ f(x) √(1 + (f'(x))^2) dx
def surface_area(x1, x2):
    return quad(lambda x: 2 * np.pi * f(x) * np.sqrt(1 + df(x)**2), x1, x2)[0]

# Calculate total volume and surface area for the entire vase
volume_total = sum(volume(a, b) for a, b in [(0, 1.7), (1.7, 4.2), (4.2, 5.31), (5.31,
8.1), (8.1, 9.2), (9.2, 12.2), (12.2, 14.3)])
area_total = sum(surface_area(a, b) for a, b in [(0, 1.7), (1.7, 4.2), (4.2, 5.31),
(5.31, 8.1), (8.1, 9.2), (9.2, 12.2), (12.2, 14.3)])

# Output the results
print(f"Total Volume: {volume_total:.4f}")
print(f"Total Surface Area: {area_total:.4f}")
```

When this python script is executed in the terminal, it gives the output as shown in the screenshot below.



Hence, using the python script I found out that.

$$\text{Optimized Volume} = \mathbf{450.58\ cm^3} \text{ and Optimized Surface area } = \mathbf{365.48 cm^2}$$